Cyber Forensic Technologies & Process
AY2026 Semester 2
Practical 02 – Storage File System

Sample Answer:

Question 1:

The original hash of the drive image is stored in the file header of the UnknownDrive.E01 file.

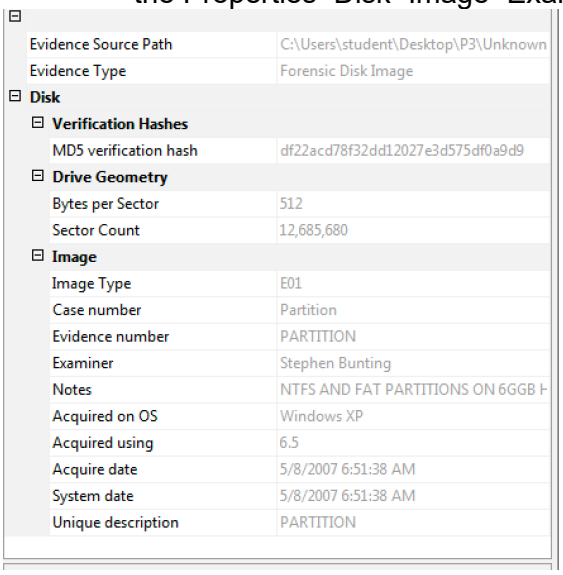
Question 2:

From the information in the Properties>Disk>DriveGeometry, Bytes per sector is 512 and Sector count is 12,685,680.

The total bytes available in the drive is 512 x 12,685,680 = 6,495,068,160 bytes (around 6GB)

Question 3:

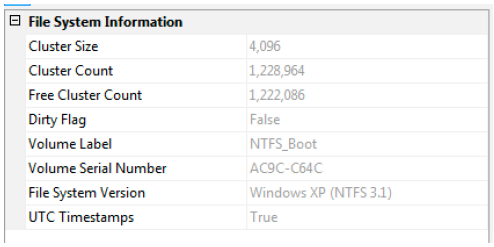
Stephen Bunting (See the screenshot of the properties, the creator of the drive image is stated in the Properties>Disk>Image>Examiner field



Evidence Source Path	C:\Users\student\Desktop\P3\Unknown
Evidence Type	Forensic Disk Image
Disk	
Verification Hashes	
MD5 verification hash	df22acd78f32dd12027e3d575df0a9d9
Drive Geometry	
Bytes per Sector	512
Sector Count	12,685,680
Image	
Image Type	E01
Case number	Partition
Evidence number	PARTITION
Examiner	Stephen Bunting
Notes	NTFS AND FAT PARTITIONS ON 6GGB H
Acquired on OS	Windows XP
Acquired using	6.5
Acquire date	5/8/2007 6:51:38 AM
System date	5/8/2007 6:51:38 AM
Unique description	PARTITION

Question 4:

From the Properties>File System Information>Cluster Size, the Cluster Size is 4,096 bytes per cluster (8 sectors/cluster)

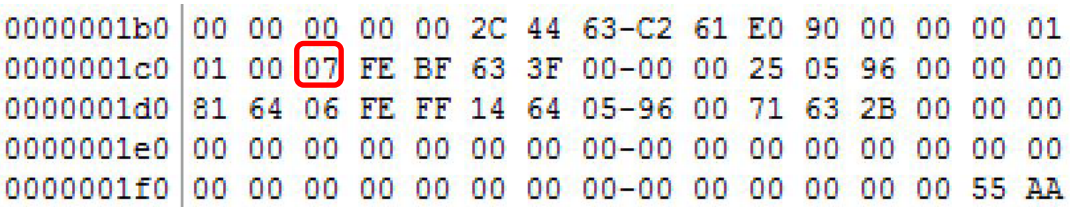


File System Information	
Cluster Size	4,096
Cluster Count	1,228,964
Free Cluster Count	1,222,086
Dirty Flag	False
Volume Label	NTFS_Boot
Volume Serial Number	AC9C-C64C
File System Version	Windows XP (NTFS 3.1)
UTC Timestamps	True

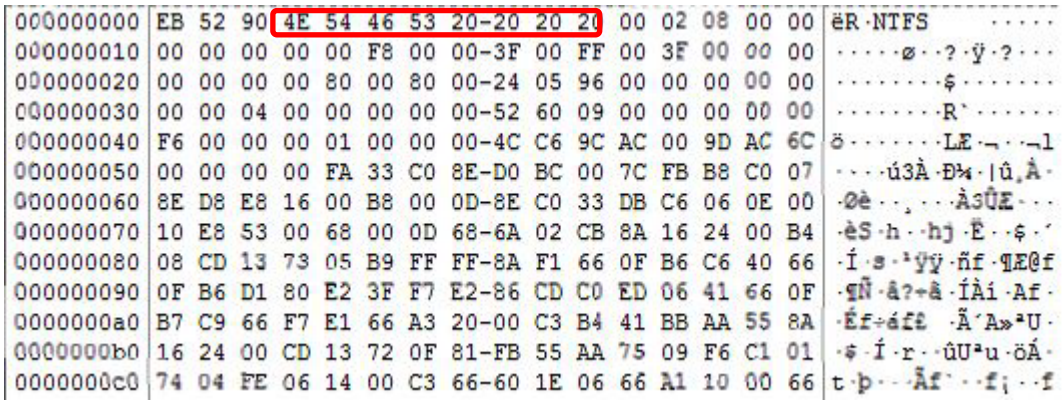
Question 5:

There are 2 ways to confirm the file system in the Partition 1 is a NTFS file system.

1) From the Master Boot Record (physical sector 0), Partition 1 entry starts from byte offset 1be (in Hex). Partition Type is at byte offset from the starting of the Partition 1 entry, that is 1be + 4 = 1c2 (in Hex). At the byte offset 1c2, it contains a value of 07. 07 means NTFS file system.



2) From the OEM ID (8 bytes) in the Partition 1 Partition Boot Sector (logical sector 0) byte offset 3, it displayed “NTFS “ that indicate the partition is in NTFS file system format.



Question 6:

1) The $MFT record that stores the properties of the file $LogFile starts at byte offset 0800 indicated by the record starting marker **FILE0**

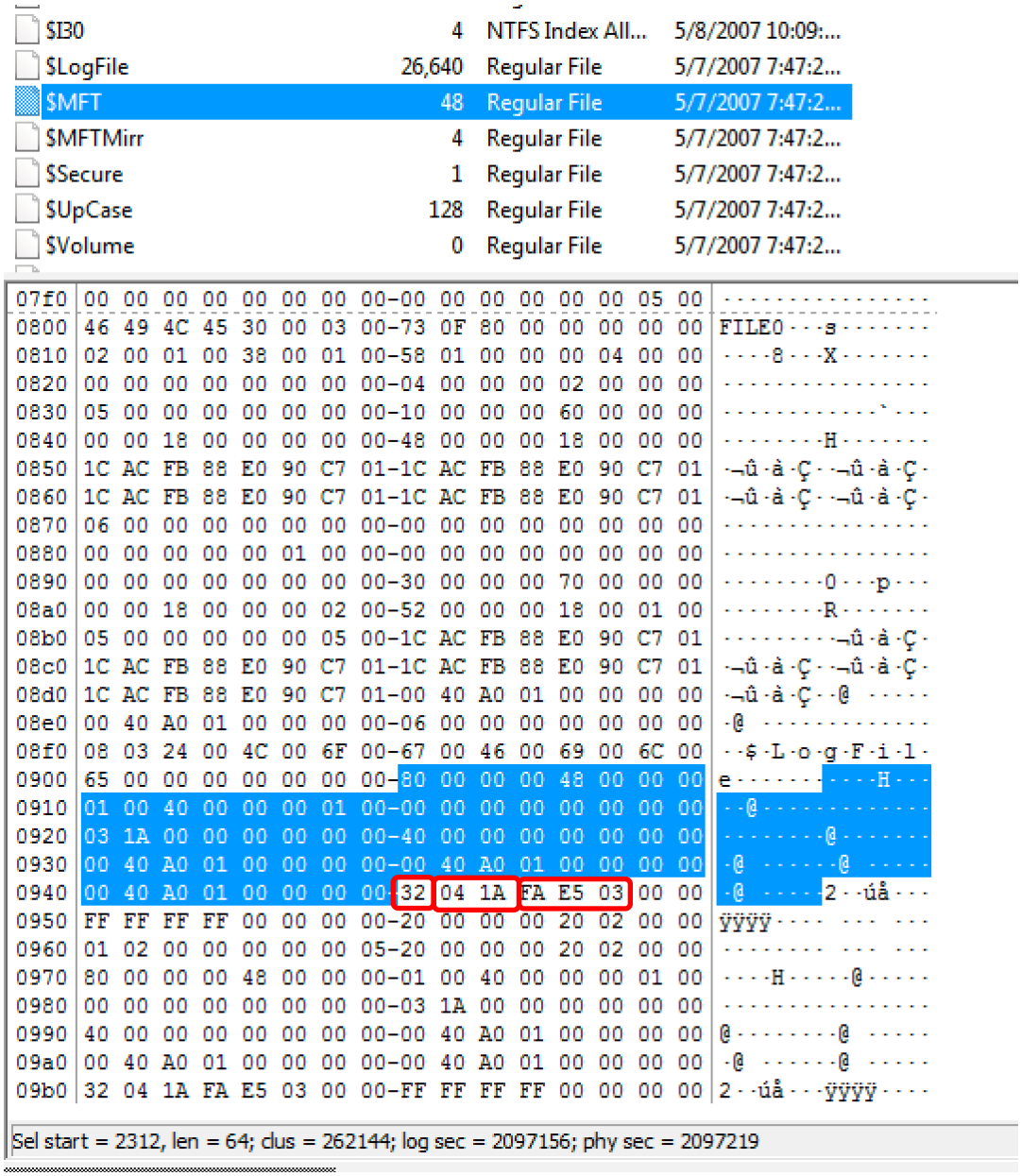
2) The starting of the $DATA section of the record is at byte offset 0x0908 indicated by the $DATA section starting marker **80 00 00 00** (4 bytes of Hex numbers)

3) The 4 bytes behind 0x80 00 00 00 (0x indicate Hex numbers) are 0x48 00 00 00 which is the length of the $DATA section. After reading these 4 bytes in little endian way, the length of the $DATA section is 0x00000048 or 72 in decimal.

4) The first Cluster Run (or called Data Run) lists the clusters used to store the content of the file $LogFile content is at byte offset 64 (in decimal) from the start of the $DATA section. It shows 32 04 1A FA E5 03 00 00

5) The digit 2 in 0x32 indicates the following 2 bytes 04 1A are the number of clusters in the Cluster Run and the digit 3 in 0x32 indicates the following 3 bytes FA E5 03 after the 2 bytes 04 1A are the starting cluster number of the Cluster Run.

6) After reading the numbers 04 1A and FA E5 03 in little endian way, the Cluster Run start at the cluster 0x03E5FA (255,482 in decimal) with a size of 0x1A04 clusters (6,660 in decimal)



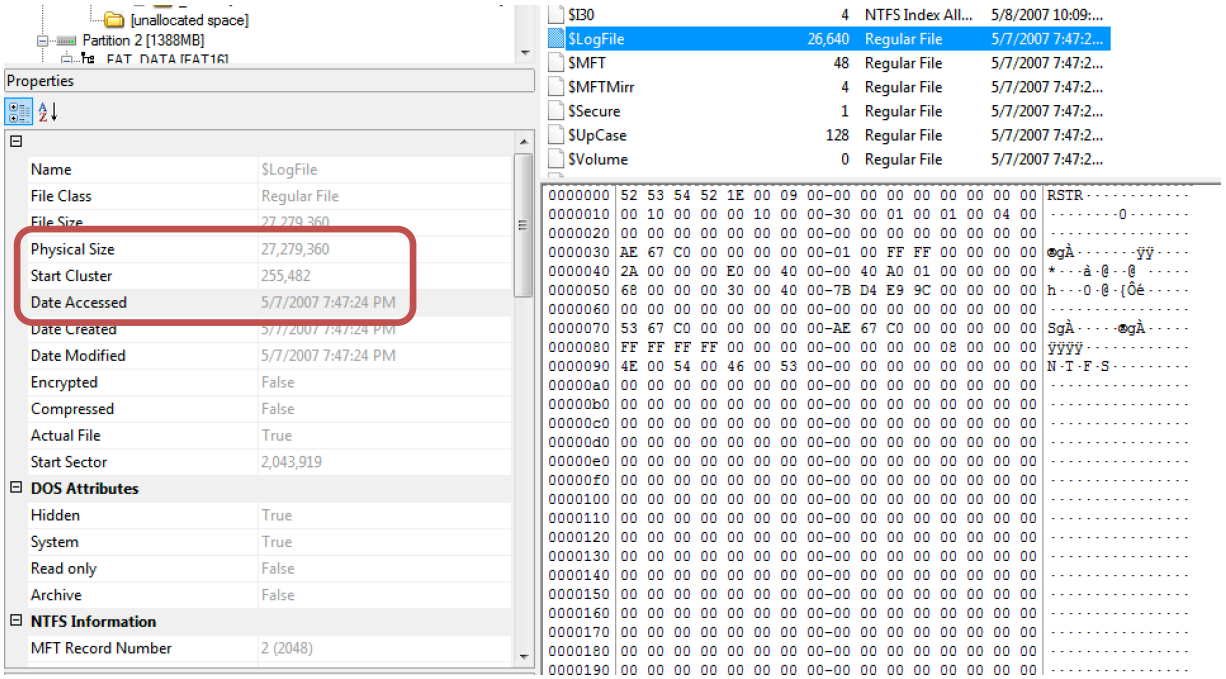
$I30	4	NTFS Index All...	5/8/2007 10:09:...
$LogFile	26,640	Regular File	5/7/2007 7:47:2...
$MFT	48	Regular File	5/7/2007 7:47:2...
$MFTMirr	4	Regular File	5/7/2007 7:47:2...
$Secure	1	Regular File	5/7/2007 7:47:2...
$UpCase	128	Regular File	5/7/2007 7:47:2...
$Volume	0	Regular File	5/7/2007 7:47:2...

```
07f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 05 00 ----------------
0800 46 49 4C 45 30 00 03 00-73 0F 80 00 00 00 00 00 FILE0---s-------
0810 02 00 01 00 38 00 01 00-58 01 00 00 00 04 00 00 ----8---X-------
0820 00 00 00 00 00 00 00 00-04 00 00 00 02 00 00 00 ----------------
0830 05 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00 ------------`---
0840 00 00 18 00 00 00 00 00-48 00 00 00 18 00 00 00 --------H-------
0850 1C AC FB 88 E0 90 C7 01-1C AC FB 88 E0 90 C7 01 -¬û-à-Ç--¬û-à-Ç-
0860 1C AC FB 88 E0 90 C7 01-1C AC FB 88 E0 90 C7 01 -¬û-à-Ç--¬û-à-Ç-
0870 06 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 ----------------
0880 00 00 00 00 00 01 00 00-00 00 00 00 00 00 00 00 ----------------
0890 00 00 00 00 00 00 00 00-30 00 00 00 70 00 00 00 --------0---p---
08a0 00 00 18 00 00 00 02 00-52 00 00 00 18 00 01 00 --------R-------
08b0 05 00 00 00 00 00 05 00-1C AC FB 88 E0 90 C7 01 ---------¬û-à-Ç-
08c0 1C AC FB 88 E0 90 C7 01-1C AC FB 88 E0 90 C7 01 -¬û-à-Ç--¬û-à-Ç-
08d0 1C AC FB 88 E0 90 C7 01-00 40 A0 01 00 00 00 00 -¬û-à-Ç--@ -----
08e0 00 40 A0 01 00 00 00 00-06 00 00 00 00 00 00 00 -@ ------------
08f0 08 03 24 00 4C 00 6F 00-67 00 46 00 69 00 6C 00 --$-L-o-g-F-i-l-
0900 65 00 00 00 00 00 00 00-80 00 00 00 48 00 00 00 e-----------H---
0910 01 00 40 00 00 00 01 00-00 00 00 00 00 00 00 00 --@-------------
0920 03 1A 00 00 00 00 00 00-40 00 00 00 00 00 00 00 --------@-------
0930 00 40 A0 01 00 00 00 00-00 40 A0 01 00 00 00 00 -@ ------@ -----
0940 00 40 A0 01 00 00 00 00-32 04 1A FA E5 03 00 00 -@ -----2--úå---
0950 FF FF FF FF 00 00 00 00-20 00 00 00 20 02 00 00 ÿÿÿÿ---- --- ---
0960 01 02 00 00 00 00 00 05-20 00 00 00 20 02 00 00 -------- --- ---
0970 80 00 00 00 48 00 00 00-01 00 40 00 00 00 01 00 ----H-----@-----
0980 00 00 00 00 00 00 00 00-03 1A 00 00 00 00 00 00 ----------------
0990 40 00 00 00 00 00 00 00-00 40 A0 01 00 00 00 00 @--------@ -----
09a0 00 40 A0 01 00 00 00 00-00 40 A0 01 00 00 00 00 -@ ------@ -----
09b0 32 04 1A FA E5 03 00 00-FF FF FF FF 00 00 00 00 2--úå---ÿÿÿÿ----
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Sel start = 2312, len = 64; clus = 262144; log sec = 2097156; phy sec = 2097219

7) The previous step, the starting cluster number for the Cluster Run for the file $LogFile is 255,482 which matches with the number Start Cluster 255,482 displayed in the Properties Pane.

The size of the file $LogFile obtained from the previous step is 6,660 clusters or 6,660 x 4096 bytes/cluster = 27,279,360 bytes which also matches with the Physical Size 27,279,360 bytes displayed in the Properties Pane.



Question 7: Complete and record the steps with relevant screenshots on the following tasks

a) Locate the File Record of the file **special_image.jpg.pgp** in the $MFT file
b) Locate the first cluster data runs
c) Examine and determine the cluster size (in number of clusters) and starting cluster number for each cluster data runs used by the file **special_image.jpg.pgp**

Sample answer:

The cluster data run is 31 36 61 58 06
The cluster size is 0x36 clusters and the starting cluster is 0x065861